BolCode: Code in your own language

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The problem

Millions of Indian students and developers think in Hindi, Gujarati or Hinglish, but every coding tool expects typed English. Beginners from Tier 2 and Tier 3 towns lose time translating their thoughts into English before they can even start.

The solution

BolCode is a VS Code extension where you edit code by *speaking*. Select a function, press a hotkey and say *"is function ko async bana do aur error handling add karo"*. BolCode transcribes the Hinglish, an LLM edits only the selected code, and you review a **diff** before accepting. It also explains code in simple Hinglish, undoes changes by voice, and replies out loud in Hindi.

How it uses Gnani AI

- **Prisma (speech to text):** transcribes Hindi-English mixed speech (en-IN), with word boosting for code terms such as async, await, useState.
- **Timbre (text to speech, beta):** speaks short Hindi confirmations and a one-line spoken summary of explanations.
- An OpenAI-compatible LLM performs the code edit. Nothing is applied until the user accepts the diff.

Results (my own test)

40 Hinglish coding commands, one speaker, one laptop and microphone:

- **91.8% code-term accuracy** (56 of 61 code terms recognised).
- Average word error rate 22.2%, mostly harmless spelling variants ("iss" for "is", "mein" for "me").
- Real misses (async heard as "asinc", await as "avaidh", undo as "ando") are fixed by a small correction layer before the LLM.
- A small sample from one voice, not a benchmark.

Who it helps

Students and junior developers who are more comfortable in Hindi, Gujarati or Hinglish; learners from smaller towns; anyone who prefers speaking to typing. It fits Gnani's focus on Indian languages and code-mixed voice.

Built today

Working extension (tested on Windows), open source, packaged release v0.1.0, setup-check command, accuracy test script, public demo video.

Next

Validate Gujarati commands, streaming transcription with Prisma's realtime API, voice accept/reject, and a browser version that explains DSA problems in Hinglish.

Links

- Code and release: github.com/DevAnas19/bolcode
- Demo on LinkedIn: linkedin.com/feed/update/urn:li:activity:7522609014883479552
- Demo on X: x.com/Anas19a07/status/2116845849057935608